Solve with us 2.2 - Not Graded

This assignment will not be graded and is only for practice.
Key Points: 1

A matrix is in row echelon form if :

• The first non-zero element (the leading entry) in a row is 1.
• The column containing the leading 1 of a row is to the right of the column containing the leading 1 of the row above it.
• Any non-zero rows are always above rows with all zeros.

A matrix is in reduced row echelon form if :

• The first non-zero element in the first row (the leading entry) is the number 1.
• All subsequent non-zero rows must also have their leading entries (i.e. first non-zero entries) as 1 and they should appear to the right of the leading entry in the previous row.
• The leading entry in each row must be the only non-zero number in its column.
• Any non-zero rows are always above rows with all zeros.

Note: Any matrix which is in reduced row echelon form is also in row echelon form.

1 point

Choose the correct options for the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$A = $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

[Hint: Find out the leading entries for each rows and observe whether the matrix is satisfying the conditions of row echelon form and reduced row echelon form.]

○ AA is in row echelon form but not in reduced row echelon form.
○ AA is in both row echelon form and reduced row echelon form.
○ AA is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.
Score: 0
Accepted Answers:
AA is in both row echelon form and reduced row echelon form.

1 point

Choose the correct options for the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$A = $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

○ AA is in row echelon form but not in reduced row echelon form.
○ AA is in both row echelon form and reduced row echelon form.
○ AA is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.
Score: 0
Accepted Answers:
AA is neither in row echelon form nor reduced row echelon form.
1 point

Choose the correct options for the matrix F = [110 010 001]F = [100110001]

○
AA is in row echelon form but not in reduced row echelon form.
○
AA is in both row echelon form and reduced row echelon form.
○
AA is neither in row echelon form nor reduced row echelon form.
No, the answer is incorrect.
Score: 0
Accepted Answers:
AA is in row echelon form but not in reduced row echelon form.

Key Points: 2

Let $Ax = b$Ax = b be a system of linear equations and suppose AA is in reduced row echelon form. Assume that for every zero row of A, A, the corresponding entry of bb is also 00 (i.e., if the $i-th$i − th row of AA is 00, then so is b_i bi.)

• If the $i-th$i − th column has the leading entry of some row, we call x_i xi a dependent variable.
• If the $i-th$i − th column does not have the leading entry of any row, we call x_i xi an independent variable.

1 point
Consider a system of linear equations:

$$\begin{aligned} 0x_1 + x_2 + 0x_3 + 0x_4 &= 1 \\ 0x_1 + 0x_2 + x_3 + 0x_4 &= 1 \end{aligned}$$ 0x1 + x2 + 0x3 + 0x4 = 10x1 + 0x2 + x3 + 0x4 = 1

Choose the set of correct options.
[Hint: Recall the definitions of independent and dependent variable with respect to reduced row echelon form.]
☐
x_1 and x_2 x1 and x2 are dependent variables.
☐
x_2 x2 is a dependent variable.
☐
x_3 and x_4 x3 and x4 are independent variables.
☐
x_4 x4 is an independent variable.
No, the answer is incorrect.
Score: 0
Accepted Answers:
x_2 x2 is a dependent variable.
x_4 x4 is an independent variable.

1 point

let $Ax = b$Ax = b be a system of linear equations, where $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$A = $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$, and

$b = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix}$b = $\begin{bmatrix} 3 & 2 & 0 & 0 \end{bmatrix}$.

Choose the set of correct options.

☐

x_1 x1 is a dependent variable.

☐

x_2 x2 is a dependent variable.

☐

x_3 x3 is a dependent variable.

☐

The expressions x_1 = 3x1 = 3 and x_3 = 2 − x_4 x3 = 2 − x4 show the dependency of x_1 x1 and x_3 x3 on the elements of bb and the independent variables.

No, the answer is incorrect.

Score: 0

Accepted Answers:

x_1 x1 is a dependent variable.

x_3 x3 is a dependent variable.

The expressions x_1 = 3x1 = 3 and x_3 = 2 − x_4 x3 = 2 − x4 show the dependency of x_1 x1 and x_3 x3 on the elements of bb and the independent variables.

Key Points: 3

The three different types of elementary row operations that can be performed on a matrix are:

- **Type 1:** Interchanging two rows.
- **Type 2:** Multiplying a row with some constant.
- **Type 3:** Adding a scalar multiple of a row to another row.

Using these row operations one can obtain row echelon form or reduced row echelon form of a given matrix.

Consider the following matrix A and answer the questions (a), (b), (c), (d), and (e).

$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 2 & 4 & 4 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 & 3 & 2 & 4 & 2 & 1 & 4 & 0 & 0 & 3 \end{bmatrix}$$

1 point

What will be the third row of the matrix if we perform the row operation $R_3 - 2R_1$ R3 − 2R1 on AA?

○

(1123)(1123)

○
(41083)(41083)

○
$(0-203)$(0−203)

○
(0443)(0443)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(0-203)$(0−203)

1 point
What will be the second row of the matrix if we perform the row operation $\frac{1}{2}R_2$ 21R2 on AA?

○
(0110)(0110)

○
(0420)(0420)

○
$(01\frac{1}{2}0)$(01210)

○
$(122\frac{3}{2})$(12223)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(01\frac{1}{2}0)$(01210)

1 point
If we perform the row operations mentioned in question (a) and question (b), simultaneously on AA, the matrix (let it be called BB) we get is:

○
$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 1 & -2 & 2 & 1 & 0 & 0 & 0 & 3 \end{bmatrix}$$

○
$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 1 & 0 & 3 & 1 & 1 & 2 & 1 & 2 & 0 & 0 & 3 \end{bmatrix}$$

○
$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 1 & 0 & 0 & 3 & 4 & -2 & 2 & 0 & 0 & 0 & 3 \end{bmatrix}$$

○
$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 4 & 4 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 1 & 0 & 0 & 3 & 1 & 4 & 2 & 2 & 1 & 4 & 0 & 0 & 3 \end{bmatrix}$$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 3 \\ -2 & 2 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

1 point
Which row operations have to be performed on the matrix BB (matrix BB is as in question (c)) to make all the entries 0 in the second column, except the leading entry of the second row.

- $R_1 - 3R_2$ R1 − 3R2 and $R_3 - 2R_2$ R3 − 2R2
- $R_1 + 3R_2$ R1 + 3R2 and $R_3 + 2R_2$ R3 + 2R2
- $R_1 + 3R_2$ R1 + 3R2 and $R_3 - 2R_2$ R3 − 2R2
- $R_1 - 3R_2$ R1 − 3R2 and $R_3 + 2R_2$ R3 + 2R2

No, the answer is incorrect.
Score: 0
Accepted Answers:
$R_1 - 3R_2$ R1 − 3R2 and $R_3 + 2R_2$ R3 + 2R2
1 point
The reduced row echelon form of AA is

[Hint: Recall the definition of reduced row echelon form.]

- $$\begin{bmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 2 & 1 & 2 & 1 & 0 & 0 & 3 \end{bmatrix}$$
- $$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 3 \end{bmatrix}$$
- $$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & -2 & 3 & 3 \end{bmatrix}$$
- $$\begin{bmatrix} 1 & 0 & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & -2 & -3 & 3 & 3 \end{bmatrix}$$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & \frac{3}{2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & -2 & 3 & -2 & 3 \end{bmatrix}$$

Key Points: 4

Let $Ax = b$Ax = b denote the matrix representation of a system of linear equations.

- The augmented matrix is denoted by $[A|b]$[A|b].
- The matrix obtained by performing the operations that transform AA to its reduced row echelon form RR on the augmented matrix is denoted by $[R|c]$[R|c].
- The solutions of $Rx = c$Rx = c are the same as the solutions of $Ax = b$Ax = b.

Consider a system of linear equations

$2x_1 + x_2 = 1 - x_1 + x_3 + x_4 = -1x_1 + x_2 - x_3 + x_4 = 2 - x_1 + x_3 + x_4 = 1$.2x1 + x2 = 1−x1 + x3 + x4 = −1x1 + x2 − x3 + x4 = 2−x1 + x3 + x4 = 1.

Answer the questions (a), (b) and (c) based on the above information.

1 point
Which of the following represents an augmented matrix of the system?

[Hint: First try to write down the coefficient matrix AA and the corresponding vector bb.]

○
○
○
○

No, the answer is incorrect.
Score: 0
Accepted Answers:

1 point
What will be the matrix $[R|c]$[R|c] obtained by performing the operations that transform AA to its reduced row echelon form RR on the augmented matrix?

○
○
○
○

No, the answer is incorrect.
Score: 0
Accepted Answers:

1 point
Choose the correct option. [Hint: Write down the system of linear equations corresponding to the reduced row echelon form of the augmented matrix.]

○ The system of linear equations has an infinite number of solutions.
○ The system of linear equations has a unique solution.
○ The system of linear equations has no solution.
○
$x_1 = x_2 = x_3 = x_4 = 1$x1 = x2 = x3 = x4 = 1 is a solution of the system of linear equations.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The system of linear equations has no solution.

Check Answers